2 Langevin dynamics and the score

Exercise 2.1 (ULA on a Gaussian target)**.** Let $U(x) = \frac{1}{2}\langle x, Ax\rangle$, where A is a symmetric $n \times n$ matrix with eigenvalues $0 < \lambda_{\min} = \lambda_1 \le \cdots \le \lambda_n = \lambda_{\max}$, so that $\mu = \mathcal{N}(0, A^{-1})$, and consider ULA with step size $h > 0$,

$$X_{k+1} = (\mathbb{I}_n - hA)X_k + \sqrt{2h}\,\xi_k, \qquad \xi_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \mathbb{I}_n),$$

with X_0 independent of $(\xi_k)_k$. Example 2.8 treats the isotropic case $A = \alpha\mathbb{I}_n$.

(a) Let $h < 2/\lambda_{\max}$. Show that X_k converges in law to $\mu_h = \mathcal{N}(0, (A(\mathbb{I}_n - hA/2))^{-1})$ whatever the law of X_0, and deduce that μ_h is the unique stationary law of the chain. Show that for $h \ge 2/\lambda_{\max}$ the variance of $\langle e_n, X_k\rangle$, where e_n is a unit eigenvector for $\lambda_{\max}$, grows at least linearly in k.

(b) Let $m_k = \mathbb{E}X_k$. Show that $|m_k| \le \rho(h)^k |m_0|$ with $\rho(h) = \max_i |1 - h\lambda_i|$, with equality for suitable m_0. Show that the best rate is $\min_h \rho(h) = \frac{\kappa-1}{\kappa+1}$, where $\kappa = \lambda_{\max}/\lambda_{\min}$ is the condition number, and deduce that reducing the mean by a constant factor takes a number of iterations of order κ.

Exercise 2.2 (Score matching for Gaussian families)**.** Example 2.12 computes the score matching objective for the scores $-\Sigma^{-1}(x-m)$ of the Gaussians $\mathcal{N}(m, \Sigma)$. Here we parametrise by the precision matrix $A = \Sigma^{-1}$ and allow any symmetric A, writing $s_{m,A}(x) = -A(x-m)$ and $J(m, A) := \mathbb{E}_\mu |s_{m,A}(X)|^2 + 2\mathbb{E}_\mu \nabla \cdot s_{m,A}(X)$. The computation in the example gives $J(m, A) = \operatorname{tr}(A \operatorname{Cov}_\mu(X) A) - 2\operatorname{tr} A + |A(\mathbb{E}_\mu X - m)|^2$ for any μ with finite second moments. Given data $x_1, \dots, x_N \in \mathbb{R}^n$ with sample mean $\hat{m}$ and sample covariance $\hat{\Sigma} = \frac{1}{N}\sum_i (x_i - \hat{m})(x_i - \hat{m})^\top$, show that the empirical objective $\hat{J}_N(m, A) = \frac{1}{N}\sum_{i=1}^N \left(|s_{m,A}(x_i)|^2 + 2\nabla \cdot s_{m,A}(x_i)\right)$ has the unique minimiser $(\hat{m}, \hat{\Sigma}^{-1})$ when $\hat{\Sigma} \succ 0$, which is the Gaussian maximum likelihood estimator. Show that $\hat{J}_N$ is unbounded below when $\hat{\Sigma}$ is singular, in particular when $N \le n$.

Exercise 2.3 (Stein's identity and denoising score matching)**.** Let $X \sim \mu$ (any probability measure on $\mathbb{R}^n$) and $Z \sim \mathcal{N}(0, \sigma^2\mathbb{I}_n)$ be independent, $Y = X + Z$ and $\nu = \operatorname{Law}(Y)$. Recall that ν has a positive C^∞ density with $\mathbb{E}_\nu |\nabla \log \nu|^2 \le n/\sigma^2$, and that $\nabla \log \nu(y) = -\sigma^{-2}\mathbb{E}[Z \mid Y = y]$ (Tweedie's formula, Proposition 2.15).

(a) (Stein's identity.) Let $F \in C^1(\mathbb{R}^n; \mathbb{R}^n)$ with $\mathbb{E}|F(Z)|^2 < \infty$ and $\mathbb{E}|\nabla \cdot F(Z)| < \infty$. Show that $\mathbb{E}\,\nabla \cdot F(Z) = \sigma^{-2}\mathbb{E}\langle F(Z), Z\rangle$. *Hint:* apply Lemma A.7 to $G = F\varphi_{\sigma^2\mathbb{I}_n}$, where $\varphi_{\sigma^2\mathbb{I}_n}$ is the density of Z.

(b) (Denoising score matching.) Let $s_\theta \in C^1(\mathbb{R}^n; \mathbb{R}^n)$ with $\mathbb{E}|s_\theta(Y)|^2 < \infty$ and $\mathbb{E}|\nabla \cdot s_\theta(Y)| < \infty$. Show that

$$\mathbb{E}|s_\theta(Y) - \nabla \log \nu(Y)|^2 = \mathbb{E}\left|s_\theta(Y) + \frac{Z}{\sigma^2}\right|^2 + C', \qquad C' = \mathbb{E}_\nu |\nabla \log \nu|^2 - \frac{n}{\sigma^2},$$

using the score matching identity $\mathbb{E}|s_\theta(Y) - \nabla \log \nu(Y)|^2 = \mathbb{E}|s_\theta(Y)|^2 + 2\mathbb{E}\,\nabla \cdot s_\theta(Y) + \mathbb{E}_\nu |\nabla \log \nu|^2$ of Proposition 2.14 together with (a). This is a second proof of Proposition 2.16.

(c) Let $\mu = \mathcal{N}(0, a^2\mathbb{I}_n)$. Compute $\nabla \log \nu$, $\mathbb{E}[X \mid Y]$ and C', and discuss the bias $\nabla \log \nu - \nabla \log \mu$ and the minimal value $-C'$ of the denoising objective as $\sigma \to 0$.

Exercise 2.4 (Weak sensitivity to well-separated mixture weights)**.** For probability densities q, p on $\mathbb{R}^n$ that are C^1 and positive on the open set $\{q > 0\}$, the Fisher divergence is $\mathrm{FI}(q\|p) := \int_{\{q>0\}} q\,|\nabla \log q - \nabla \log p|^2 \,\mathrm{d}x$. If q and p are positive and C^1 on all of $\mathbb{R}^n$, then $\mathrm{FI}(q\|p) = 0$ implies $q = p$, since $\log(q/p)$ then has zero gradient on the connected set $\mathbb{R}^n$, and both densities integrate to one. Fix a unit vector $e \in \mathbb{R}^n$. Only the coordinate $\langle e, x\rangle$, in the direction separating the modes, will matter.

(a) Let $\rho(x) = c\, e^{-1/(1-|x|^2)}$ for $|x| < 1$ and $\rho(x) = 0$ otherwise (a C^∞ probability density on $\mathbb{R}^n$, c the normalising constant), and for $w \in (0,1)$ let $q_w(x) = w\rho(x + 2e) + (1 - w)\rho(x - 2e)$. Show that $\nabla \log q_w$ does not depend on w on $\{q_w > 0\}$, hence the Fisher divergence (computed on $\{q_w > 0\}$) satisfies $\mathrm{FI}(q_w\|q_{w'}) = 0$ for all w, w', although $q_w \neq q_{w'}$ for $w \neq w'$. Why does this not contradict the implication $\mathrm{FI}(q\|p) = 0 \Rightarrow q = p$ stated above? What does the Langevin dynamics $\mathrm{d}X_t = \nabla \log q_w(X_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t$ do when started at $X_0 = -2e$?

(b) Let $q_w = w\mathcal{N}(-me, \mathbb{I}_n) + (1 - w)\mathcal{N}(me, \mathbb{I}_n)$, $m > 0$. Show that $\nabla \log q_w(x) = -x + m \tanh(m\langle e, x\rangle + c_w)\, e$ with $c_w = \frac{1}{2} \log \frac{1-w}{w}$.

(c) Show that $\mathrm{FI}(q_w\|q_{1/2}) > 0$ for $w \neq \frac{1}{2}$, but that $\mathrm{FI}(q_w\|q_{1/2}) \leq C_w m^2 e^{-m^2/8}$ for a constant C_w depending only on w, so that $\mathrm{FI}(q_w\|q_{1/2}) \to 0$ as $m \to \infty$. *Hint:* $\tanh' = \mathrm{sech}^2 \leq 4e^{-2|\cdot|}$; split according to $|X| \leq m/2$ or not.